

Consider the Circuit shown in the figure below:

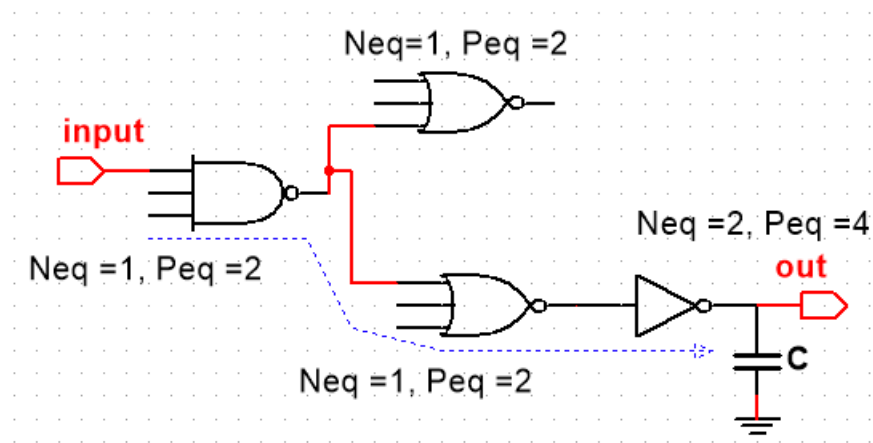


Figure 1

First: Use the Microwind Layout tool to do the following

1. Create the **NAND3_hw2.MSK** gate as follows

- Create a copy of the layout **A_Book_NAND3.MSK** and save it under another name, say **NAND3_hw2.MSK**. Put it in a folder call it hw2.
- Keep the N diffusion as is and change the P such that the Ratio of $P_{eq}:N_{eq} = 2$.
- Add Metal1 with a Metal Contact to the first Input Poly Input. See Figure 2 below
- Keep nearly everything the same. Specially the distance between the Vdd and Ground. Note that you will need to make some changes to the contacts of the P-diffusion.

e. Simulate the gate by using a clock at the first input and obtain the No load delay and make sure it works. Put Vdd on the other 2 inputs. Find the delay between the first input and the output when there is no load

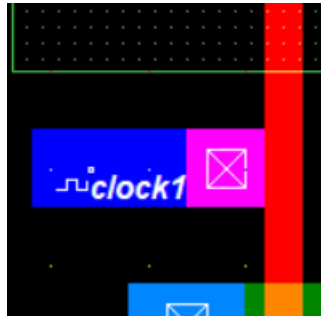


Figure 2

2. Create the **NOR3_hw2.MSK** gate as follows

- Create a copy of the layout **A_Book_NOR3.MSK** and save it under another name, save say **NOR3_hw2.MSK**. Put it in a folder call it hw2.
- Reduce the N diffusion to make $N_{eq} = 1$. Increase the P size the $P_{eq} = 2$. The Ratio of $P_{eq}:N_{eq} = 2$.
- Add Metal1 with a Metal Contact to the first Input Poly Input. See Figure 2
- Keep nearly the other things the same. Specially the distance between the Vdd and Ground. Note that you will need to make some changes to the contacts of the P and N diffusion.
- Simulate the gate by using a clock input and obtain the No load delay and make sure it works. Put Vss(GND) on the other 2 inputs. Find the delay between the first input and the output when there is no load (No Capacitor)

3. Create the **INV_hw2.MSK** gate as follows

- Create a copy of the layout **A_Book_NOR2.MSK** and save it under another name, save say **INV_hw2.MSK**. Put it in a folder call it hw2.
- Cut the Second input. Cut vertically all at the start of the second vertical poly to convert to an inverter and change the connections.

c. Reduce the N diffusion to make $N = 2$ ($W = 8l$). Keep the P size = $16L$ which makes $P = 4$. Notice that Ratio of $P_{eq}:N_{eq} = 2$. But the Input and out capacitance are more than the unit inverter. Notice that p and g do not change.

d. Add Metal1 with a Metal Contact to the first Input Poly Input as shown in Figure 2.

Keep nearly the other things the same. Specially the distance between the Vdd and Ground. Note that you will need to make some changes to the contacts of the P and N diffusion.

e. Simulate the gate by using a clock input and obtain the No load delay and make sure it works. Put Vss(GND) on the other 2 inputs. Find the delay between the first input and the output when there is no load (No Capacitor)

4. Create the layout for the Circuit shown above call it **hw2.MSK** in Figure1. Use insert Layout to get the layouts of the NAND3, NOR3 and inverter. Connect all Vdds and Grounds together. You will need to use Metal to connect the output NAND3 gates to one of the two NOR3 gates.(It may differ depending on your layout, but you will need to use Metal 2 for one of the connections).

5. Simulate the Circuit and find out the delay between the Input and the output (out). Make sure you change the display to show the delay between the input and the output out (Not between the input and the output of the other NOR3)

Do two simulations:

a. Simulation without the C load (NO load)

b. Simulation with a C of **0.02pF**. You can just put a capacitor as shown in Figure 3

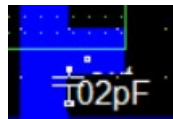


Figure3

Second: Use the Logical Effort model to compute manually the delay of the circuit in Figure 1.

Use $C = 21$ units.